Wuqian Tang

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Education

National Tsing Hua University *Hsinchu, Taiwan*
Ph.D. Candidate in Computer Science (GPA: 4.25/4.3) *Sep. 2023 – Jul. 2027 (Expected)*
Supervisors: Prof. Chun-Yao Wang and Prof. Bei Yu (The Chinese University of Hong Kong)

National Tsing Hua University *Hsinchu, Taiwan*
M.S. in Computer Science (GPA: 4.11/4.3) *Sep. 2021 – Jul. 2023*
Supervisor: Prof. Chun-Yao Wang

Ming Chuan University *Taoyuan, Taiwan*
B.S. in Computer Science and Information Engineering *Sep. 2017 – Jun. 2021*

Selected Awards and Honors

Honorable Mention (Prob. A) in ICCAD 2026 CAD Contest *Nov. 2026*

Second Place in MLCAD 2026 Contest *Sep. 2026*

Excellent Teaching Assistant Award, National Tsing Hua University *Aug. 2026*

Outstanding Teaching Assistant Award (3×), National Tsing Hua University *Aug. 2025, Feb. 2026, Aug. 2026*

Distinguished Alumnus Award, Department of CSIE, Ming Chuan University *Jun. 2026*

2nd Place in IWLS 2026 Programming Contest *May 2026*

Hsinchu City Outstanding Youth (College/University Division) *Mar. 2026*

1st Place Award (Prob. C) & 2nd Place Award (Prob. A) in ICCAD 2025 CAD Contest *Oct. 2025*

First Place in ICCAD 2025 CADathlon Programming Contest *Oct. 2025*

Presidential Hsing Chien Award, NTHU (1 of 13 university-wide recipients for holistic excellence) *May 2025*

2nd Place Award (Prob. A) & Honorable Mention (Prob. B) in ICCAD 2024 CAD Contest *Oct. 2024*

Second Place in ICCAD 2024 CADathlon Programming Contest *Oct. 2024*

Synopsys - Prof. Chung Laung Liu Ph.D. Scholarship *Sep. 2024 – Present*

“Qilin Cup” (1st among 400+ teams) in Integrated Circuit EDA Elite Challenge *Dec. 2023*

Honorable Mention (Prob. A) in ICCAD 2023 CAD Contest *Oct. 2023*

Third Place in 2023 EDAthon *Aug. 2023*

1st Place Award (Prob. A) in ICCAD 2022 CAD Contest *Oct. 2022*

National Tsing Hua University Mainland China Student Scholarship *Sep. 2021 – Present*

Silver Award in International Collegiate Programming Contest (ICPC Regional) *Nov. 2020*

Work Experience

Giga Design Automation *Shanghai, Mainland China*
R&D Intern in Logic Synthesis Group (SYN), PI: Dr. Yun Shao *Jun. 2025 – Sep. 2025*

Shenzhen Research Institute, The Chinese University of Hong Kong *Shenzhen, Mainland China*
Assistant Researcher in Design Automation Laboratory, PI: Prof. Bei Yu *Jan. 2024 – Mar. 2024*

Teaching Experience

Teaching Assistant, National Tsing Hua University *Hsinchu, Taiwan*

- Introduction to Programming (Prof. Yu-Guang Chen, Department of Electrical Engineering) *Fall 2026*
- Hardware Security (Profs. Yu-Guang Chen and TingTing Hwang, Institute of Information Security) *Spring 2026*
- Advanced Logic Synthesis (Prof. TingTing Hwang, Department of Computer Science) *Spring 2025, Spring 2026*
- Introduction to Integrated Circuit Design (Prof. Yu-Guang Chen, Department of Computer Science) *Fall 2025*
- Computer Architecture (Prof. TingTing Hwang, Department of Computer Science) *Fall 2024, Fall 2025*
- Design Automation (I) (Prof. Jeong-Tyng Li, College of Semiconductor Research) *Spring 2025*

Technical Skills

Programming: Verilog, C/C++, CUDA, Python, Java, SQL, and MATLAB

Certification: Expert Level in Collegiate Programming Examination (certified by ACM-ICPC Taiwan Council)

Language: Fluent in English, Native in Mandarin (Proficient in Traditional and Simplified Chinese)

Publications

An asterisk (*) indicates equal contribution.

Conference Papers

[C19] Yu-Cheng He, Han-Wen Tsao, Yung-Chih Chen, Yi-Ting Li, **Wuqian Tang**, Jian-Meng Yang, Chun-Yao Wang, "Observability-Aware Pattern Generation for Hardware Trojan Detection", *IEEE International Conference on Computer Design (**ICCD**)*, Hong Kong SAR, China, Nov. 16–18, 2026 *(to appear)*.

[C18] Bo-Ying Wang, Yong-Fong Chang, Hsun-Wei Pao, Yung-Chih Chen, Tien-Hsin Peng, Yu-Ting Hung, Tsai-Chieh Cheng, Yi-Ting Li, **Wuqian Tang**, Chun-Yao Wang, Shih-Chieh Chang, "A Design Rule-Aware Post-Route ECO Framework for Fixing Electromigration Violations", *ACM/IEEE International Conference on Computer-Aided Design (**ICCAD**)*, San Jose, California, USA, Nov. 8–12, 2026 *(to appear)*.

[C17] **Wuqian Tang***, Zixiao Wang*, Che-Chien Lin, Fong-Hua Fu, Laura Angélica Shion-Korsak, Xinyun Zhang, Bei Yu, Chun-Yao Wang, "TAPCO: LLM-Guided Design-Adaptive Timing and Power Co-Optimization", *ACM/IEEE International Symposium on Machine Learning for CAD (**MLCAD**)*, Jeju, South Korea, Sep. 7–9, 2026.

[C16] Tao-Chun Huang, Hao-Yu Tsai, You-Cheng Lin, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Jian-Meng Yang, Chun-Yao Wang, "Late Breaking Results: QUBO Approaches to Logic Equivalence Checking and Testing", *ACM/IEEE Design Automation Conference (**DAC**)*, Long Beach, California, USA, Jul. 26–29, 2026.

[C15] Fangzhou Liu*, **Wuqian Tang***, Bo-Ying Wang, An-Chieh Shen, Han-Wen Tsao, Yuyang Ye, Yun Shao, Chun-Yao Wang, Bei Yu, "HyPAS: A Hybrid Optimization Framework for Placement and Sizing Co-Optimization", *ACM/IEEE Design Automation Conference (**DAC**)*, Long Beach, California, USA, Jul. 26–29, 2026.

[C14] Chi-Wei Chen, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Jian-Meng Yang, Chun-Yao Wang, "Learning to Approximate: Circuit Learning and Deep Reinforcement Learning for Approximate Logic Synthesis with an Error Rate Guarantee", *IEEE/ACM Design, Automation and Test in Europe (**DATE**)*, Verona, Italy, Apr. 20–22, 2026.

[C13] You-Cheng Lin, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Chia-Chieh Chu, Chun-Yao Wang, "A Mathematical Exploration to Equivalence Checking of Quantum Circuits", *IEEE/ACM Design, Automation and Test in Europe (**DATE**)*, Verona, Italy, Apr. 20–22, 2026.

[C12] Zi-Wei Huang, Yi-Ting Li, **Wuqian Tang**, Liang-Chia Cheng, Yung-Chih Chen, Chun-Yao Wang, "Process Selection Strategy for Chiplet-Based 2.5D Systems under Design Constraints", *IEEE International VLSI Symposium on Technology, Systems and Applications (**VLSI-TSA**)*, Hsinchu, Taiwan, Apr. 13–17, 2026.

[C11] Yi-Ting Shen, Yan-Hsiu Liu, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Hao-Chiang Shao, Chia-Wen Lin, Chun-Yao Wang, "Integrating Automatic Prompt Engineering and Vision-Language Model for Pad Defect Classification", *IEEE International Symposium on Quality Electronic Design (**ISQED**)*, San Francisco, California, USA, Apr. 8–10, 2026.

[C10] Kai-Po Hsu, Yi-Ting Li, Fu-Cheng Cai, Chia-Feng Chien, **Wuqian Tang**, Ting-Yu Ku, Yu-Chen Cheng, Tsung-Han Lai, Cheng-Lung Wang, Yi-Ting Shen, Kuan-Ling Chou, Zi-Wei Huang, Tao-Chun Huang, Tzu-Li Hsu, Yung-Chih Chen, Shih-Chieh Chang, Ting-Chi Wang, TingTing Hwang, Chun-Yao Wang, "An Efficient Approach to NP3-Boolean Matching: Leveraging Multi-bit Ports and Reverse Engineering to Minimize Search Space", *IEEE International Symposium on Quality Electronic Design (**ISQED**)*, San Francisco, California, USA, Apr. 8–10, 2026.

[C9] Yu-Chuan Yen, Fu-Cheng Cai, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Ihao Chen, Chun-Yao Wang, "Area-Oriented Threshold Logic Circuit Synthesis Using Negative Weights", *IEEE International Symposium on Quality Electronic Design (**ISQED**)*, San Francisco, California, USA, Apr. 8–10, 2026.

[C8] Chia-Feng Chien, Ming-Wei Chen, Yin-Chen Chang, Yung-Chih Chen, Yan-Hsiu Liu, Hao-Chiang Shao, Chia-Wen Lin, Yi-Ting Li, **Wuqian Tang**, Chun-Yao Wang, "Improving CNN-based Pad Defect Classification with Enhanced Preprocessing Techniques", *IEEE International SoC Design Conference (**ISOCC**)*, Busan, South Korea, Oct. 15–18, 2025.

[C7] Kuan-Ling Chou, Cheng-Lung Wang, Yung-Chih Chen, **Wuqian Tang**, Yi-Ting Li, Shih-Chieh Chang, Chun-Yao Wang, "CNN Model Optimization Using a Hybrid Approach of Genetic Algorithm-based Pruning and Retraining with Knowledge Distillation", *ACM Great Lakes Symposium on VLSI (**GLSVLSI**)*, New Orleans, Louisiana, USA, Jun. 30 – Jul. 2, 2025.

[C6] Yu-Che Lee, Yu-Hsuan Chen, Yu-Chen Cheng, Yong-Fong Chang, Jia Wei Lin, Hsun-Wei Pao, Yung-Chih Chen, Yi-Ting Li, **Wuqian Tang**, Shih-Chieh Chang, Chun-Yao Wang, "Real-Time Dynamic IR-drop Prediction for IR ECO", *ACM/IEEE Design Automation Conference (**DAC**)*, San Francisco, California, USA, Jun. 22–25, 2025.

[C5] Kai-Po Hsu, Tsung-Han Lai, Yi-Ting Li, **Wuqian Tang**, Yun-Ju Lee, Yung-Chih Chen, Wen-Hsin Chiu, Chun-Yao Wang, "IMU-based Motion Trajectory Reconstruction and Recognition with Dynamic Calibration on Embedded System Platform", *IEEE International Symposium on Quality Electronic Design (**ISQED**)*, San Francisco, California, USA, Apr. 23–25, 2025.

[C4] Yu-Cheng Lee, Kai-Po Hsu, Yun-Ju Lee, Yi-Ting Li, **Wuqian Tang**, Yung-Chih Chen, Wen-Hsin Chiu, Chun-Yao Wang, "On Construction of Trajectory of Boxer's Punch using a single IMU", *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (**ISPACS**)*, Kaohsiung, Taiwan, Dec. 10–13, 2024.

[C3] **Wuqian Tang**, Chuan-Shun Huang, Yung-Chih Chen, Yi-Ting Li, Shih-Chieh Chang, Chun-Yao Wang, "Model Reduction Using a Hybrid Approach of Genetic Algorithm and Rule-based Method", *IEEE International System-on-Chip Conference (**SOCC**)*, Dresden, Germany, Sep. 16–19, 2024.

[C2] Chuan-Shun Huang, **Wuqian Tang**, Yung-Chih Chen, Yi-Ting Li, Shih-Chieh Chang, Chun-Yao Wang, "An Efficient Approach to Iterative Network Pruning", *IEEE International VLSI Symposium on Technology, Systems and Applications (**VLSI-TSA**)*, Hsinchu, Taiwan, Apr. 22–25, 2024.

[C1] **Wuqian Tang**, Yi-Ting Li, Kai-Po Hsu, Kuan-Ling Chou, You-Cheng Lin, Chia-Feng Chien, Tzu-Li Hsu, Yung-Chih Chen, Ting-Chi Wang, Shih-Chieh Chang, TingTing Hwang, Chun-Yao Wang, "A Hybrid Approach to Reverse Engineering on Combinational Circuits", *IEEE/ACM Design, Automation and Test in Europe (**DATE**)*, Valencia, Spain, Mar. 25–27, 2024.

Journal Papers

[J2] Fangzhou Liu, **Wuqian Tang**, Zehua Pei, Ziyang Yu, Haisheng Zheng, Zhuolun He, Mengjia Dai, Tinghuan Chen, Bei Yu, "CB-EVO: Contextual Bandit Tuning with Evolutionary Search for Logic Synthesis", *ACM Transactions on Design Automation of Electronic Systems (**TODAES**)*, vol. 31, no. 6, pp. 119:1–119:26, 2026.

[J1] Yuh-Pyng Shieh, Shao-Hang Hsu, **Wuqian Tang**, Wei-Ying Ke, Po-sheng Ou, Yu-wei Chang, Ming-chorng Hwang, "The Lunar-Month-Interval Algorithm for Dating Oracle Bone Inscriptions", *Journal of Digital Archives and Digital Humanities (**DADH**)*, vol. 15, pp. 1–35, Apr. 2025 (in Chinese).

Preprints

[P3] **Wuqian Tang**, Zixiao Wang, Che-Chien Lin, Fong-Hua Fu, Laura Angélica Shion-Korsak, Xinyun Zhang, Bei Yu, Chun-Yao Wang, "TAPCO-G: LLM-Guided Design-Adaptive Timing and Power Co-Optimization with Graph-Based Physical Priors", *arXiv preprint* arXiv:2611.13931, 2026.

[P2] **Wuqian Tang**, Fangzhou Liu, Ruijie Wang, TingTing Hwang, Bei Yu, Chun-Yao Wang, "RTLREX: LLM-Guided Framework for Exact High-Level RTL Recovery", *arXiv preprint* arXiv:2611.10798, 2026.

[P1] Shih-Ying Yeh*, Tzu-Sian Wang*, Xuehai Wang, Jia-Hua Lee, Daniel Z. Kaplan, Ming-Qi Xu, **Wuqian Tang**, Chun-Yao Wang, Shang-Hong Lai, Chun-Yi Lee, "Trinity: One Differentiable Physics for Training, Refining and Scoring Generative Floorplanners", *arXiv preprint* arXiv:2610.04957, 2026.